

Translated

From: e-maxx.ru

Finding the largest zero submatrix

You are given a matrix with n rows and m columns. Find the largest submatrix consisting of only zeros (a submatrix is a rectangular area of the matrix).

Algorithm

Elements of the matrix will be $a[i][j]$, where $i = 0 \dots n - 1$, $j = 0 \dots m - 1$. For simplicity, we will consider all non-zero elements equal to 1.

Step 1: Auxiliary dynamic

First, we calculate the following auxiliary matrix: $d[i][j]$, nearest row that has a 1 above $a[i][j]$. Formally speaking, $d[i][j]$ is the largest row number (from 0 to $i - 1$), in which there is a element equal to 1 in the j -th column. While iterating from top-left to bottom-right, when we stand in row i , we know the values from the previous row, so, it is enough to update just the elements with value 1. We can save the values in a simple array $d[i]$, $i = 1 \dots m - 1$, because in the further algorithm we will process the matrix one row at a time and only need the values of the current row.

```
vector<int> d(m, -1);
for (int i = 0; i < n; ++i) {
    for (int j = 0; j < m; ++j) {
        if (a[i][j] == 1) {
            d[j] = i;
        }
    }
}
```

Step 2: Problem solving

We can solve the problem in $O(n \cdot m^2)$ iterating through rows, considering every possible left and right columns for a submatrix. The bottom of the rectangle will be the current row, and using $d[i][j]$ we can find the top row. However, it is possible to go further and significantly improve the complexity of the solution.